

Fluid Flow Through Pipes

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Executive Summary

For this experiment, the objective was to determine the energy loss coefficients on various sections of a piping system and compare the data to the available literature. The piping system consisted of copper, stainless steel, and plastic pipes and copper fittings. Water was used as the fluid pumped through the pipes. To ensure accurate flowrate measurements, calibration of the rotameter was performed and the flowrates were adjusted according to the experimentally determined rates of about 8, 12, 16, 24, and 31 LPM. Flow coefficients, C_f , for the orifice meter were calculated over a range of 0.59 to 0.70, showing a variance according to the flow rate and pressure differential. An inverse relationship with the calculated Reynold's number, Re , was also revealed with C_f decreasing as Re increased. Friction factor, f , and head loss for the pipes agreed with the literature, showing a decrease in f and increase in head loss as the Reynolds number, Re , increased. At a certain Re , f becomes steadier and at increasing Re , head loss increases at an accelerated rate. This was true for all pipe materials. Loss coefficient, K , of the fittings decreased with the increase in flow which effected the total head loss of the system. Compared to the pipe head loss, the fitting head loss was higher. This effected the turbulence of the flow which in turn increased the friction inside the piping.

Introduction

The overall objective of this experiment is to use the Hampden H-6920 Pipe Friction Demonstrator to measure the pressure drop of water flowing throughout the 3 various diameters of the different materials of piping that include copper pipes, stainless steel pipes, and plastic pipes. The most important part of chemical engineering is the analysis of these flow systems. It is very important to calculate the pressure drop in various sections to have a strong decision in selecting a pump that will produce the desired flow that is needed. Moreover, the significance in this lab is to obtain energy loss coefficients such as friction factor, f , loss coefficient, K , and flow coefficient, C_f , as a function of Reynolds number on various parts of the fluid-flow system including a copper 180° bend, a copper 90° Elbow bend, a section of copper pipe with an expansion and then a contraction, and the orifice plate of the flowmeter and to use these data to test the and then compare this data to available theories and correlations of fluid flowing through pipes.

This experiment showcases important parts of chemical processes that involve pipe systems which can be key in transporting materials to and from various reactors, separation processes, and storage units involving single-phase fluids and solids or two-phase flows. [1] Additionally, this experiment displays the uses of fluid flow in pipe systems in the industries where there is a need to understand the fluid flow in pipes of different material and diameters such as hot and cold water that is pumped into neighborhoods, the transportation of crude oil from upstream to downstream facilities to refine it, and the cooling of and heating of areas all over the united states to aid in providing comforting temperatures for human beings to live in. Most industrial applications are in the oil and gas industry, plumbing industries, thermal heating and cooling industries, and more.

The experimental apparatus consists of mounted pipe system, pump, reservoir tank, and a rotameter to determine the flow rates. There are three different pipes consisting of PVC, copper,

and stainless steel material with three different diameter types; 3/8", 1/2", and 1" to assist in obtaining various pressure differentials for different types of diameters and materials. After the reservoir was filled up with enough water and the flowrate was calibrated, there were three different flowrates that were used starting from 10 standard cubic feet per minute to 30 standard cubic feet per minute used to obtain the pressure drops as obtained on the DPT display. Using the tubes and various flowrates, the differential pressure was obtained in the pipes and other parts of the fluid-flow system that included expansion, contraction, elbows, and an orifice. There was always at least one valve to allow the circular flow of water in the system as the different pipes were being measured.

Theory

Fluid flow in systems is usually carried out to allow a better comprehension of pipe systems and transportation of various components to and from desired locations. In this experiment, the pipes have a desired flowrate that is set to determine the pressure differential at different flow rates ran through them. When the inlet and outlet tubing was presented around the orifice, a drop in pressure is experienced. This change can also be used to measure the flowrate of the fluid as shown in Figure 1 below.

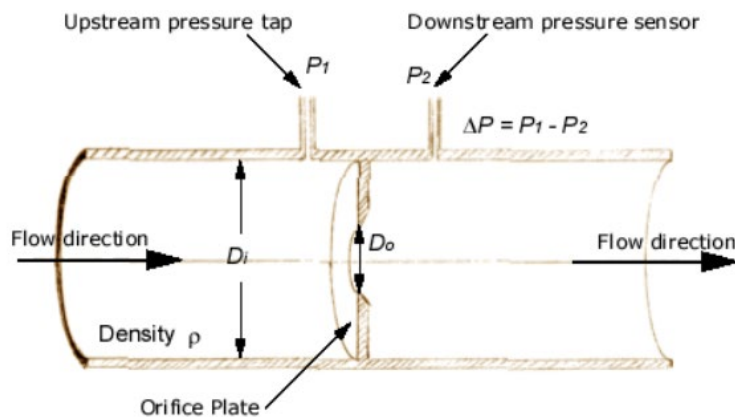


Figure 1: Orifice Flow Diagram Displaying Pressure Differential [1]

When the fluid speed is subsonic, ($V < \text{mach } 0.3$), the incompressible Bernoulli's equation describes the flow reasonably well. The experimental data will have to be determined to see if it meets this criteria. To get the pressure differential equation to a stream with this tubing is below.

$$\Delta p = p_1 - p_2 = \frac{1}{2} \rho V_2^2 - \frac{1}{2} \rho V_1^2 \quad [1]$$

The subscript of 1 describes the upstream of the orifice and the 2 is behind the orifice. Additionally, from continuity, the velocities can be replaced by cross-sectional areas of the flow and the volumetric flowrate Q using the equation below.

$$Q = \frac{\frac{\sqrt{2\Delta P}}{\sqrt{\rho}} A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} [1]$$

However, the equation above only applies for laminar, inviscid flows. For real flows (such as water or air) viscosity and turbulence are expected that introduce a discharge effect, C_d as presented in the equation below.

$$Q = \frac{C_d \frac{\sqrt{2\Delta P}}{\sqrt{\rho}} A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} [1]$$

When the value of A_2 is introduced to be uncertain, the flow coefficient substitution C_f is made as shown below.

$$C_f A_o = \frac{C_d A_2}{\sqrt{1 - \left(\frac{A_2}{A_1}\right)^2}} \text{ where } A_o \text{ is the area of the orifice} [1]$$

Thus, for real gases the equation below can be used.

$$Q = C_f A_o \frac{\sqrt{2\Delta P}}{\sqrt{\rho}} [1]$$

The C_f value is found from tabulated data and is frequently defined using the equation below as the ratio of the orifice diameter to the inlet diameter.

$$\beta = \frac{D_o}{D_{inlet}} [1]$$

Mass flow rate is found of the different pipe systems using the equation below.

$$Q_{mass} = \rho Q [1]$$

In order to get the friction factor in straight pipes, it is found using the pressure drop as shown below.

$$f = \frac{\Delta P D}{2L\rho V^2} [1]$$

Then the head loss can be calculated using the equation below.

$$h_L = \frac{\Delta P}{\rho} = f \left(\frac{L}{D}\right) \left(\frac{V^2}{2}\right) [1]$$

If the flow is laminar, the equation below is used:

$$f = \frac{64}{Re} [1]$$

The head loss due to pipe fittings is mainly caused by flow separation and additional turbulence that is expected usually from swirling. From this, the empirical data should be used as an estimate only. The head loss is given by the equation below.

$$h_l = K \left(\frac{V^2}{2} \right) [1]$$

Where the loss coefficient, K, is represented below.

$$K = \frac{L}{D} [1]$$

The experimental data will be used in calculating all of these calculations along with looking at theoretical data to compare results.

Results

Calibration of Rotameter

The rotameter is calibrated by setting the pump at 10 standard cubic feet per minute (SCFM) and timing how long it takes to fill a graduated cylinder with units of mL with water from the tap. The volume dispensed is divided by the time to find the calibrated flow rate in liters per minute (LPM). This is repeated for flow rates of 20 SCFM and 30 SCFM set on the rotameter. The flow rates on the rotameter are then graphed vs the calibrated flow rates in LPM. A best-fit linear trendline is added in Excel with corresponding linear equation. See Figure 1. The R^2 value of 0.9996 indicates a very good linear fit.

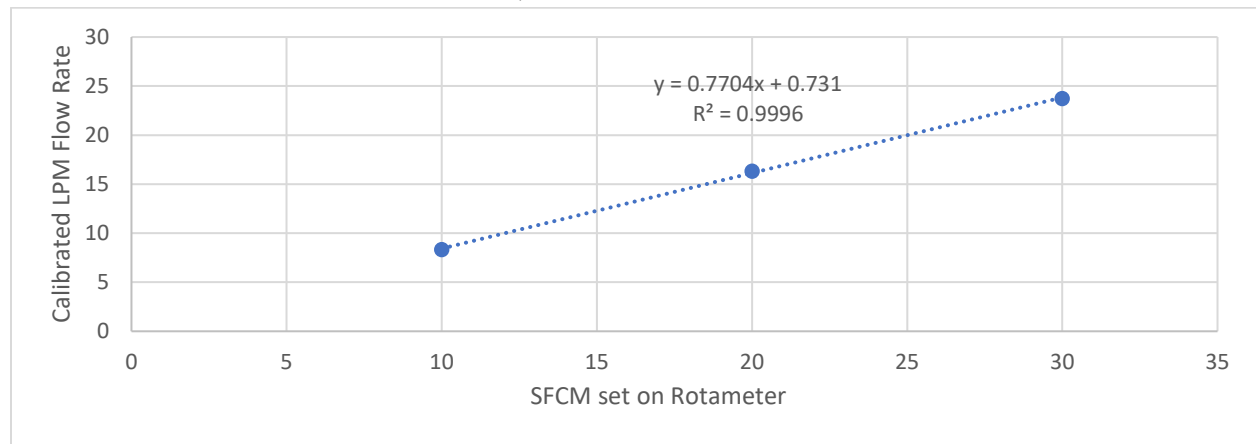


Figure 1: Calibration of Rotameter from SCFM to LPM

The equation can be used to find the conversion from SCFM on the rotameter to the calibrated flow rates in LPM. For the purpose of this experiment, flow rates of 10, 15, 20, 30, and 40 SCFM were used. See Table 1 for conversions.

Table 1: Calibrated Flow Rates

Calibrated Flow Rates				
SCFM	LPM	m ³ /s	ft ³ /s	
10	8.435	0.000141	0.004965	
15	12.287	0.0002048	0.007237	
20	16.139	0.000269	0.009506	
30	23.843	0.0003974	0.014044	
40	31.547	0.0005258	0.018581	

Orifice Meter:

Using the measured variables in Table 2, the flow coefficient, C_f , for the orifice meter was calculated for each flow rate of 8.3 LPM, 16.3 LPM, and 23.7 LPM. Beta values and mass flow rates were also determined for each flowrate. Because only one orifice was measured, the same orifice area and pipe diameter were used for each calculation. Reynolds numbers were calculated at the orifice. For the lowest flowrate of 8.3 LPM, the C_f is 0.67 and the Reynolds number is 15,0005. The medium flowrate of 16.3 LPM produced a C_f of 0.72 and Reynolds number of 30,186. At the highest flowrate of 23.7 LPM, a C_f of 0.65 and Reynolds number of 44,265 were determined (Table 3). Beta values are the same due to the ratio of orifice diameter to pipe diameter remaining unchanged.

Table 2: Summary of variables used in calculating C_f for calibrated flows.

Flowrate (LPM)	Flowrate (m ³ /s)	Velocity (m/s)	Differential Pressure (Pa)	ρ (kg/m ³)	D_i pipe Diameter (m)	D_o Orifice Diameter (m)	Orifice Area (m ²)
8.435	0.000141	1.214	1585.795	998	0.026035	0.0121412	0.0001158
16.139	0.000269	2.323	5377.913	998	0.026035	0.0121412	0.0001158
23.843	0.000397	3.432	13789.520	998	0.026035	0.0121412	0.0001158

Table 3: Calculated C_f , beta, Reynolds number, and mass flow rate for each measured flow rate.

Flow Coefficient C_f	Beta	T (F)	μ (Pa*s)	Re	Qmass (kg/s)
0.681157	0.466341	70.3	0.00097	15171.58	0.140302
0.707711	0.466341	72.4	0.000943	29847.00	0.268445
0.652938	0.466341	73	0.000936	44438.49	0.396589

Figure 2 shows the flow coefficient results plotted according to the calculated Reynolds number. The pattern of the coefficients appears to generally decrease with the increase in Reynolds number. Even though the Reynolds numbers are increasing at 12 LPM and 16 LPM, the coefficients reveal outlier behavior compared to rest of the data. At 12 LPM, C_f decreases, increases at 16 LPM, and then decreases at 23 LPM and 31 LPM.

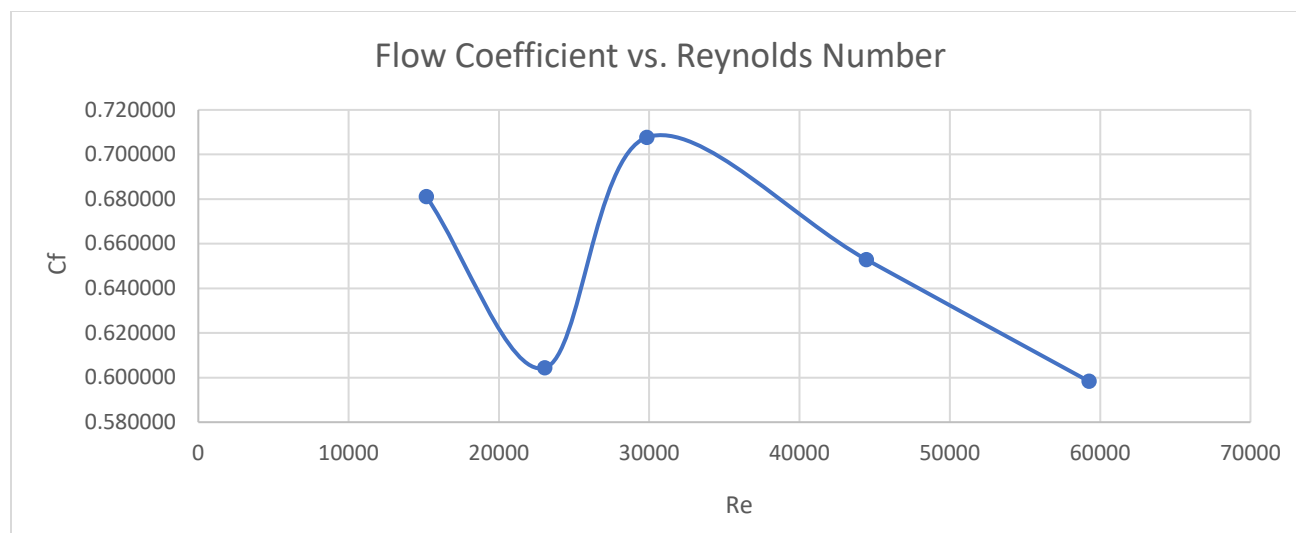


Figure 2: Flow coefficients, C_f , versus Reynolds number for calibrated flow rates.

Pipe Friction Head Loss

The Fanning friction factor (f) is calculated for 9 parallel pipes made of either copper tubing, plastic pipe, and stainless steel pipe using water as the fluid. The relative roughness (k_s), inner diameter, length, and size of each pipe is known. The Reynolds number (Re) can be calculated for each pipe at each flow rate with known values from the literature for the density and viscosity of water at room temperature. The pressure drop along each pipe is measured experimentally and recorded at 3 different flow rates. The experimental fanning friction factor is then calculated and compared to the theoretical friction factor obtained from the Colebrook Equation. The Darcy friction factor is also calculated for comparison to the fanning friction factor. The Darcy friction factor is used in the Moody diagrams, but is not compared to Moody values in this report. See Table 4 and Figure 3 for copper Pipes. See Appendix I for plastic and stainless steel data.

Table 4: Copper Pipe Friction Factor with Error Analysis

COPPER PIPE					
Inner Diameter (ft)	Re	f Experimental (Fanning)	f Experimental (Darcy)	f Colebrook	%Error (Colebrook vs Fanning)
0.0335	43093.6219	0.000162475	0.000649901	0.0055221	97.05772646
0.0555	26011.46547	0.000841734	0.003366937	0.006128917	86.26618276
0.085416667	16901.1083	0.00165185	0.006607398	0.006775469	75.62014435
0.0335	82512.22865	0.000602051	0.000602051	0.004857229	87.60505115
0.0555	49804.67855	0.000542681	0.002170726	0.005308554	89.77722641
0.085416667	32360.89358	0.000720908	0.002883632	0.005809998	87.59194058
0.085416667	47808.46308	0.000536741	0.002146963	0.005327617	89.92531137
0.0335	62818.49888	0.000209185	0.00083674	0.005116527	95.91158448
0.0555	37917.4723	0.0006662	0.002664802	0.005630185	88.16734274

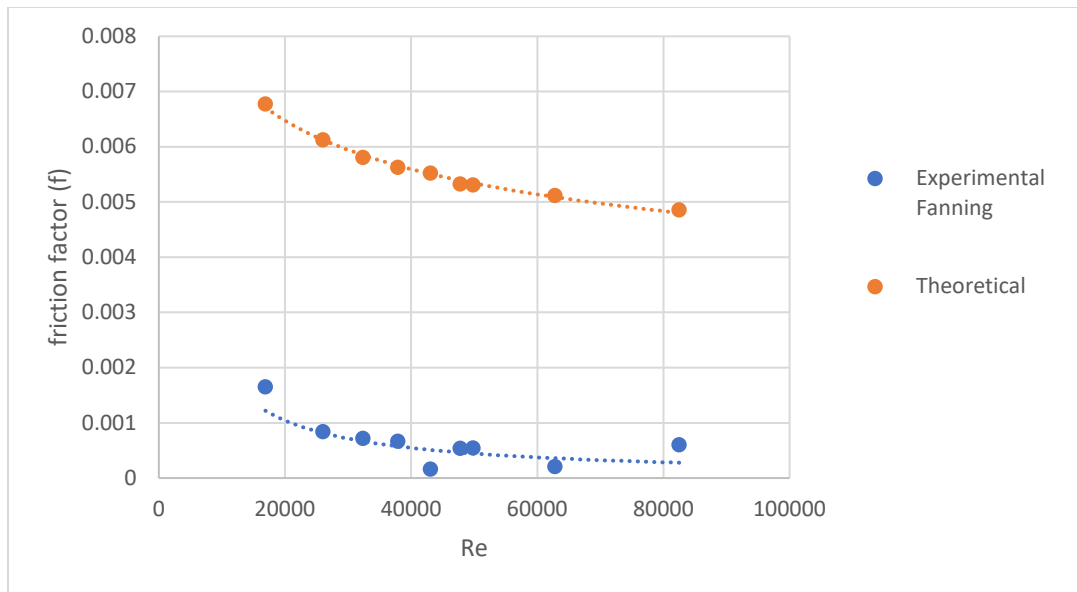


Figure 3: Reynolds number vs fanning friction factor for copper pipes comparing theoretical values obtained from the Colebrook Equation to experimental values

The errors are consistently high between the theoretical values and the experimentally obtained values for friction factor in the range of 60% to 99%. However, the experimental data seems to follow the same trend as the theoretical values in the graphs with a decrease in friction factor as the Reynolds number increases. The friction factor values seem to level out towards the higher Reynolds values. There is a consistent difference between the theoretical values and experimental values of about 0.004 – 0.006.

The head loss due to friction is calculated and graphed vs pressure drop for all pipes and flow rates and graphed vs Reynolds number. Again, large errors are present but the general trends in the graphs are the same. As the Reynolds number increases, the head loss increases at an increasing rate. The errors between theoretical and experimental head loss are smaller at lower Reynolds numbers. See Table 5 and Figure 4. See Appendix I for plastic and stainless steel pipe data.

Table 5: Head loss for copper pipes

COPPER PIPE					
Inner Diameter (ft)	Re		Head Loss Experimental (ft)	Head Loss Theoretical (ft)	% Error
0.0335	43093.6219		0.009616935	0.326853888	97.05772646
0.0555	26011.46547		0.003991935	0.029066467	86.26618276
0.085416667	16901.1083		0.000907258	0.003721343	75.62014435
0.0335	82512.22865		0.130645161	1.054019366	87.60505115
0.0555	49804.67855		0.009435484	0.092298668	89.77722641
0.085416667	32360.89358		0.001451613	0.011698952	87.59194058
0.085416667	47808.46308		0.002358871	0.023413835	89.92531137
0.0335	62818.49888		0.026310484	0.643537423	95.91158448
0.0555	37917.4723		0.00671371	0.056738816	88.16734274

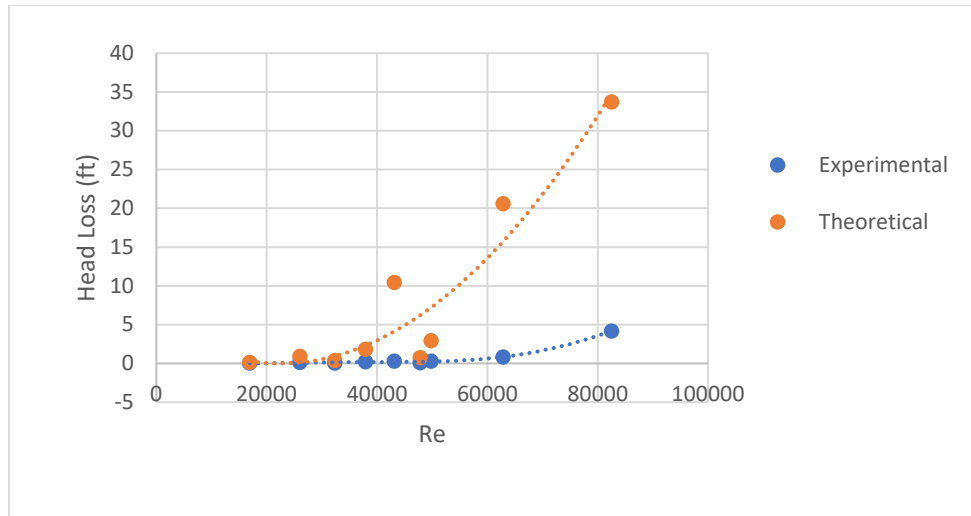


Figure 4: Copper pipes head loss vs Reynolds number using theoretical and experimental values for comparison.

The Fanning friction factor and the Darcy friction factor are calculated using the following equations:

$$\text{Darcy } f = \frac{\Delta P D^2}{L \rho V^2}$$

$$\text{Fanning } f = \frac{\Delta P D}{2 L \rho V^2}$$

The literature indicates that the Fanning friction factor is $\frac{1}{4}$ that of the Darcy friction factor used in the Moody diagrams [2,3]. This is very close to the experimental friction factors obtained in Tables 2, 3 and 4. For example, the Fanning friction factor for the $\frac{1}{2}$ inch copper pipe calculated using experimental pressure drop is 0.000162 and the Darcy friction factor is 0.000689.

Dividing the two values gives a factor of 4.

Pipe Fitting Head Loss:

Piping systems consist of multiple bends, contractions, expansions, tees, and couplings where the streams flow through sudden increase or decrease of conduit, or sharp corners and stream breakage. Such conduit disturbances in fitting and valves cause friction that can be much higher than the friction loss in straight pipes. The pipe fittings included in this system are two 90° elbows, T-bend (180° degree elbow), expansion and contraction. The pipe fittings also cause additional turbulence that increases energy loss, or friction loss that is calculated with equation 5.65. [3]

$$h_L = K * \frac{V^2}{2 * g} \quad (5.65)$$

From the lab manual, (equation 5.8 in McCabe, Smith, Hariott,):

$$f = \frac{\Delta P D}{2L\rho V^2}$$

$$h_L = \frac{\Delta P}{\rho} = K \frac{V^2}{2g}$$

where loss coefficient K is:

$$K = f * \frac{L}{D}$$

However since we don't know exact length of all the pipe fittings we used equality above to compute values of loss coefficient for each fittings:

$$K = \frac{\Delta P}{\rho} * \frac{2g}{V^2}$$

Table 6 includes data calculated using the equations above, and values are calculated for the copper tubing system.

Density of water is 62.4 lbs/ft³ and the gravity is 32.174 lb/fts.

Velocity V is calculated from the flow rate:

$$V = \frac{Q}{A}$$

$$A = \pi * r^2 = \pi * \frac{D^2}{4}$$

Table 6: Detailed results for the calculations of K loss coefficient

	dP(psi)			Velocity (ft/s)			D (ft)	A (ft ²)	dP(lb/ ft ²)			K, loss coefficient		
Q(ft ³ /s) =	0.004964627	0.009499	0.0140334	0.0049646	0.009499	0.014033			0.004965	0.009499	0.014033	0.004965	0.009499	0.014033
Elbow 1(left)	0.06	0.06	0.08	0.8660549	1.657055	2.448055	0.085433	0.005732	8.64	8.64	11.52	11.87882	3.244818	1.98226
elbow2 (90)	0.07	0.08	0.12	0.8660549	1.657055	2.448055	0.085433	0.005732	10.08	11.52	17.28	13.85863	4.326424	2.973389
U bend	0.35	0.06	0.1	0.8660549	1.657055	2.448055	0.085433	0.005732	50.4	8.64	14.4	69.29313	3.244818	2.477824
Expansion	0.06	0.06	0.06	0.157188	0.3007537	0.444319	0.200534	0.031584	8.64	8.64	8.64	360.5998	98.50139	45.13088
Contraction	0.39	0.07	0.1	0.157188	0.3007537	0.444319	0.200534	0.031584	56.16	10.08	14.4	2343.898	114.9183	75.21814

Table 7: Details from table 6

	ΔP (psi)			K, loss coefficient		
Q(ft ³ /s) =	0.0049646	0.009499	0.014033	0.0049646	0.009499	0.014033
Elbow 1(left)	0.06	0.06	0.08	11.87882	3.244818	1.98226
elbow2 (90)	0.07	0.08	0.12	13.85863	4.326424	2.973389
T bend	0.35	0.06	0.1	69.29313	3.244818	2.477824
Expansion	0.06	0.06	0.06	360.5998	98.50139	45.13088
Contraction	0.39	0.07	0.1	2343.898	114.9183	75.21814

After calculations of the K, loss coefficient total head loss of the copper pipe system is calculated including straight copper pipe size of 1-1/8" and inner diameter of the 1.025", and well as all fittings listed above. Total head loss is calculated according do the textbook equation 5.68. [3]

$$h_f = \left(\frac{4fL}{D} * V_{spipe}^2 + K_{elbow1} * V_{e1}^2 + K_{elbow2} * V_{e2}^2 + K_{expansion} * V_{expans}^2 + K_{cont} * V_{cont}^2 \right) * \frac{1}{2g}$$

The $\frac{4fL}{D} * \frac{V^2}{2g}$ is a skin friction loss in the straight tube that is calculated in the previous section of the pipe friction head loss.

After computation of the total head loss for flow rates of 10, 20 and 30 SCFM the results are following:

Q(ft ³ /s)	h _L (ft)
0.0049646	2.149763
0.009499	0.767314
0.0140334	1.070923

Discussion

Calibration of Rotameter:

The Rotameter in the Unit Ops laboratory stated that the conversion from SCFM is 1 SCFM is 8.45 LPM. This is consistent or very close the values obtained experimentally to calibrate the rotameter. When the rotameter is set to 10 SCFM, the calibrated flow rate is 8.435 LPM. The difference could be that the rotameter could have been set a little bit above or below the actual 10 SCFM line depending on the angle the operator is looking at the line.

Orifice Meter:

Flow coefficients vary according to the flow rate and generally decrease with the increase in Reynolds number. Two coefficient outliers deviate from the pattern, but show an increase in Reynolds number with flow rate as expected. These deviations could be explained by the coefficient not being valid for the Reynolds number. According to McCabe, Smith, and Harriot, a coefficient of 0.61 is not accurate when Re at the orifice is less than 30,000. [3] For flowrates of 16 LPM and below, Re is less than 30,000 and different coefficients may be valid for one or all of the flowrates.

Reynolds number calculations are consistent with the relationship between the viscosity and expected Reynolds number. The higher the viscosity, the lower the Reynolds number is

expected. Table 3 shows the increase in temperature for each flowrate. This corresponds to a change in the water viscosity, which is dependent on the temperature.

Pipe Friction Head Loss and Friction Factor:

The Darcy and Fanning friction factors agree with the literature and are different by a factor of 4. The Fanning friction factor was compared with the theoretical value calculated using the Colebrook equation and a difference of about 0.004 to 0.006 is consistently noted. This may be due to the various versions of Colebrook available and the correct version is not utilized. The graphs of Re vs Fanning friction factor show similar trends with a decrease in f as Re increases, which becomes steadier as Re number increases. The trends appear to resemble the Moody diagram, which utilized the Darcy friction factor.

Head loss is also calculated and the general trend is noted that as the Reynolds number increases, the head loss will increase at an accelerated rate.

These trends are consistent with all pipe materials. See Appendix I.

Pipe Fitting Head Loss:

Pipe fitting head loss always occurs due to the fittings and valves disturbance to the fluid flow and increase of the friction, as well as the turbulence. The friction loss is usually higher than the straight pipe which also agrees with the experimental data in this report. The only result that is showing errors are due to the malfunctioning apparatus and human error, is the total head loss per 20 SCFM flow rate where head loss is 0.77 ft, lower than the flow rates of the 10 SCFM and 30 SCFM. In practice where pipes are not as long, but there are many contractions and fittings with the high values of the head losses then the friction losses of the pipes don't have to be taken into consideration. [3]



References

- [1] Zdunek, Alan. "ChE 381: Unit Operations Laboratory FLUID FLOW IN PIPE SYSTEMS." *University of Illinois at Chicago, 2019.*, 26 Oct. 2019.
- [2] "Fanning Friction Factor." https://en.wikipedia.org/wiki/Fanning_friction_factor.
- [3] W. McCabe, J. Smith, and P. Harriot, Unit Operations of Chemical Engineering, 5th ed., McGraw-Hill, 1993.
- [4] "Water – Dynamic and Kinematic Viscosity Calculator." Engineering Toolbox. https://www.engineeringtoolbox.com/water-dynamic-kinematic-viscosity-d_596.html

Appendix I: Data Tabulation and

Rotameter Calibration

Table 8: Raw Data for Rotameter Calibration

Rotameter Calibration			
SCFM	Time (min)	Volume (mL)	LPM
30	0.036	855	23.75
20	0.0465	759	16.32258065
10	0.0836667	6988	8.342629

Orifice Meter:

Table 9: Summary of variables used in calculating C_f for 12.3 and 31.5 LPM.

Flowrate (LPM)	Flowrate (m ³ /s)	Velocity (m/s)	Differential Pressure (Pa)	ρ (kg/m ³)	D _i pipe Diameter (m)	D _o Orifice Diameter (m)	Orifice Area (m ²)
12.287	0.000205	1.769	4274.751	998	0.026035	0.0121412	0.0001158
31.547	0.000526	4.541	28751.149	998	0.026035	0.0121412	0.0001158

Table 10: Calculated C_f , beta, Reynolds number, and mass flow rate for 12.3 and 31.5 LPM.

Flow Coefficient C_f	Beta	T (F)	μ (Pa*s)	Re	Qmass (kg/s)
0.604333	0.466341	73.5	0.000930	23050.7	0.204374
0.598296	0.466341	73.6	0.000929	59259.39	0.524732

Pipe Friction Head Loss for Straight Pipes

Table 11: Raw Data and Calculations for friction factors and head loss

COPPER PIPE	Inner Diameter (ft)	Area (ft ²)	V (ft/s)	Re	ΔP (ftH ₂ O)	f Experimental (Fanning)	f Experimental (Darcy)	f Colebrook	%Error (Colebrook vs Fanning)	Head Loss Experimental (ft)	Head Loss Theoretical (ft)	Length (ft)
0.035	0.00081413	5.632676518	43093.6219	76.32	0.00062475	0.00064901	0.0055221	87.05772646	0.00961635	0.32695388	4	
0.055	0.002419223	2.052577719	2601.46547	31.68	0.00041734	0.00236337	0.00652917	86.2461276	0.00399195	0.023066447	4	
0.08546667	0.00573027	0.865386099	1601.003	7.2	0.00059585	0.00607298	0.00577649	75.6209435	0.00907259	0.00372143	4	
0.035	0.00081413	10.78480889	82512.2285	259.2	0.00062091	0.000602051	0.004857229	87.6050515	0.10045161	1.054019366	4	
0.055	0.002419223	3.523307967	48804.67855	74.88	0.000542681	0.002170726	0.005308954	89.77722641	0.009435484	0.032288688	4	
0.08546667	0.00573027	1.688888372	32360.85358	11.52	0.000720946	0.002683832	0.005693986	87.59184058	0.01451613	0.01636952	4	
0.08546667	0.00573027	2.45071637	47685.46239	18.72	0.000536741	0.002446863	0.005327617	89.32531037	0.00225871	0.023413825	4	
0.035	0.00081413	8.210727854	62888.49888	208.8	0.000209895	0.0003674	0.00516527	96.9188448	0.026310484	0.643507423	4	
0.055	0.002419223	2.891474502	37917.4723	53.28	0.0006662	0.002684802	0.00563085	88.16734274	0.00671371	0.05673886	4	
PLASTIC PIPE												
0.041083333	0.001225627	3.7451681	35139.221	106.56	0.000629295	0.002517141	0.005659395	89.8807004	0.012427419	0.12077779	4	
0.051833333	0.002101025	2.352784383	27851.50483	36	0.000679631	0.002178523	0.005756554	88.6286707	0.00453629	0.038865334	4	
0.08746667	0.006001756	0.827195627	18594.42898	10.89	0.00258631	0.010385242	0.006786508	61.74305542	0.001270161	0.00320089	4	
0.041083333	0.001225627	7.170850834	67281.77671	328.32	0.000528599	0.002114536	0.004833525	89.19267569	0.04170968	0.362804907	4	
0.051833333	0.002101025	4.504885607	53237.8391	82.16	0.000414572	0.001696339	0.00541943	90.76347001	0.01161863	0.12600018	4	
0.08746667	0.006001756	1.503848193	31620.59088	12.96	0.000910522	0.003642088	0.005798994	84.30057334	0.001633065	0.004023395	4	
0.08746667	0.006001756	2.339028589	46714.65649	18.72	0.000602591	0.002410365	0.005303422	88.63769189	0.002598871	0.020760491	4	
0.041083333	0.001225627	5.4593727	51223.18787	208.8	0.000580276	0.002321104	0.005195372	88.83080548	0.026310484	0.259565057	4	
0.051833333	0.002101025	3.429675211	40599.73722	53.28	0.000473354	0.001895344	0.005474254	91.35309454	0.00671371	0.07764298	4	
Stainless Steel Pipe												
0.041083333	0.001225627	3.7451681	35139.221	41.76	0.000246812	0.000988447	0.007685314	96.78275742	0.005262097	0.163593247	4	
0.051833333	0.002101025	2.352784383	27851.50483	24.48	0.000462148	0.001848586	0.007507927	93.84451839	0.0013084577	0.05016288	4	
0.08746667	0.006001756	0.827195627	18594.42898	7.24	0.000845089	0.00746803	0.007568874	75.49823699	0.006801258	0.003762628	4	
0.041083333	0.001225627	7.170850834	67281.77671	233.28	0.000375759	0.000502873	0.007234595	94.8767404	0.02333981	0.512788886	4	
0.051833333	0.002101025	4.504885607	53237.8391	50.4	0.000259532	0.001038127	0.00767457	96.32779185	0.006550806	0.172942442	4	
0.08746667	0.006001756	1.503848193	31620.59088	10.88	0.000708194	0.002823735	0.006830708	89.63235171	0.001270161	0.0022592	4	
0.08746667	0.006001756	2.339028589	46714.65649	11.52	0.000770625	0.004453201	0.006597229	84.30352092	0.001451613	0.024462634	4	
0.041083333	0.001225627	5.4593727	51223.18787	80.64	0.000224107	0.000839426	0.007452442	96.38284315	0.01161863	0.137903627	4	
0.051833333	0.002101025	3.429675211	40599.73722	36	0.000319534	0.001727334	0.007226734	95.5743009	0.00453629	0.102498842	4	

Table 12: Plastic Pipe Friction Factor with Error Analysis

PLASTIC PIPE						
Inner Diameter (ft)	Re	f Experimental (Fanning)	f Experimental (Darcy)	f Colebrook	%Error (Colebrook vs Fanning)	
0.041083333	35139.2211	0.000629285	0.002517141	0.005659395	88.8807004	
0.051833333	27851.50483	0.000679631	0.002718523	0.005975654	88.6266707	
0.087416667	16514.42898	0.002596311	0.010385242	0.006786508	61.74305542	
0.041083333	67281.77671	0.000528859	0.002115436	0.004893525	89.19267569	
0.051833333	53327.8391	0.000474572	0.001898289	0.005149143	90.78347001	
0.087416667	31620.51088	0.000910522	0.003642088	0.005799894	84.30105734	
0.087416667	46714.65649	0.000602591	0.002410365	0.005303422	88.63769199	
0.041083333	51223.19787	0.000580276	0.002321104	0.005195372	88.83090548	
0.051833333	40599.73722	0.000473354	0.001893414	0.005474254	91.35309454	

Table 13: Stainless Steel Friction Factor with Error Analysis

Stainless Steel Pipe						
Inner Diameter (ft)	Re	f Experimental (Fanning)	f Experimental (Darcy)	f Colebrook	%Error (Colebrook vs Fanning)	
0.041083333	35139.2211	0.000246612	0.000986447	0.007665314	96.78275742	
0.051833333	27851.50483	0.000462149	0.001848596	0.007507927	93.84451939	
0.087416667	16514.42898	0.001854508	0.00741803	0.007568874	75.49823699	
0.041083333	67281.77671	0.000375768	0.001503073	0.007334555	94.8767404	
0.051833333	53327.8391	0.000259532	0.001038127	0.007067457	96.32779185	
0.087416667	31620.51088	0.000708184	0.002832735	0.006830708	89.63235171	
0.087416667	46714.65649	0.000370825	0.001483301	0.006509729	94.30352092	
0.041083333	51223.19787	0.000224107	0.000896426	0.007452442	96.99284313	
0.051833333	40599.73722	0.000319834	0.001279334	0.007226734	95.5743009	

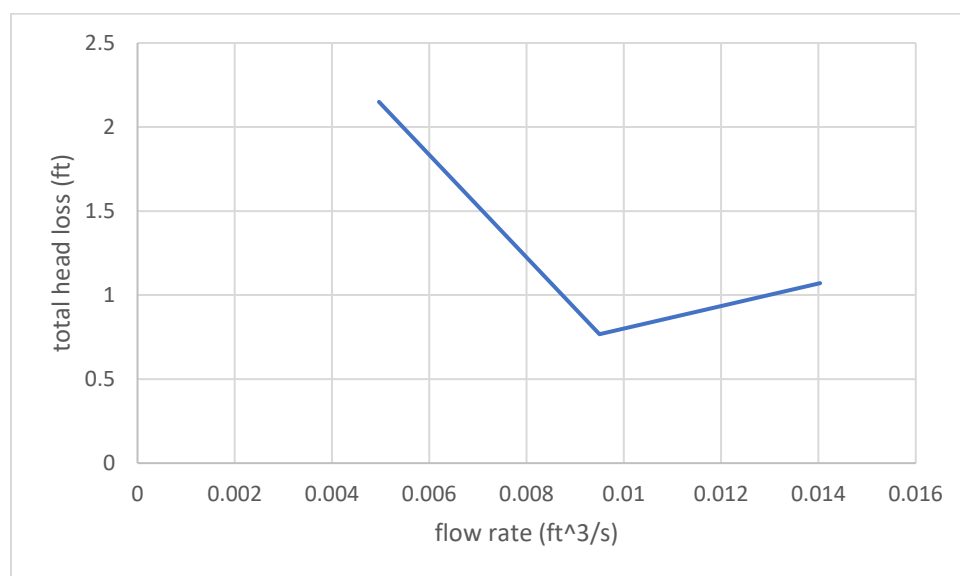


Figure 5: Graph of total head loss vs the flow rate

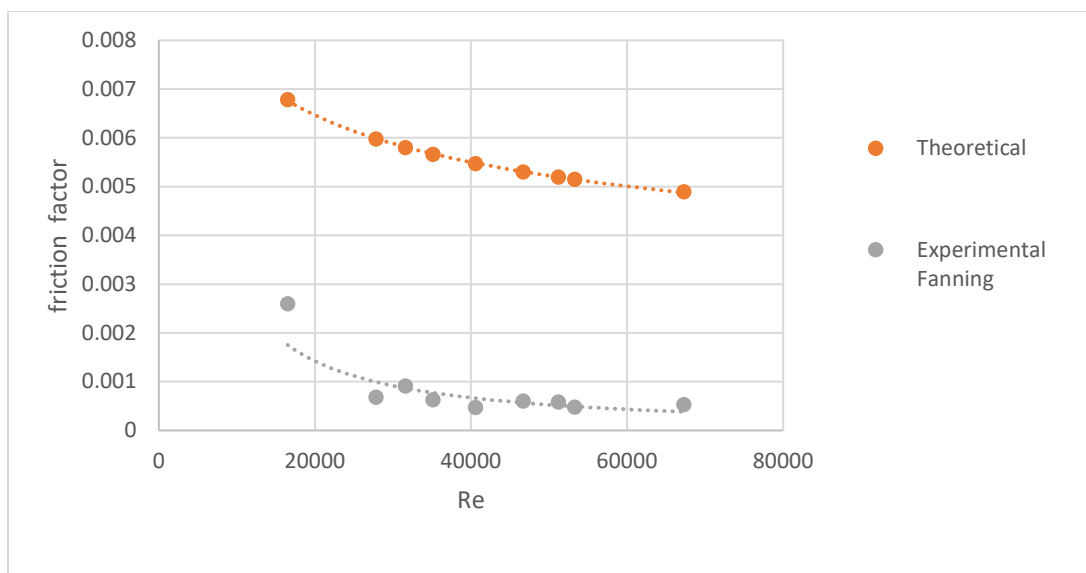


Figure 6: Reynolds number vs fanning friction factor for plastic pipes comparing theoretical values obtained from the Colebrook Equation to experimental values

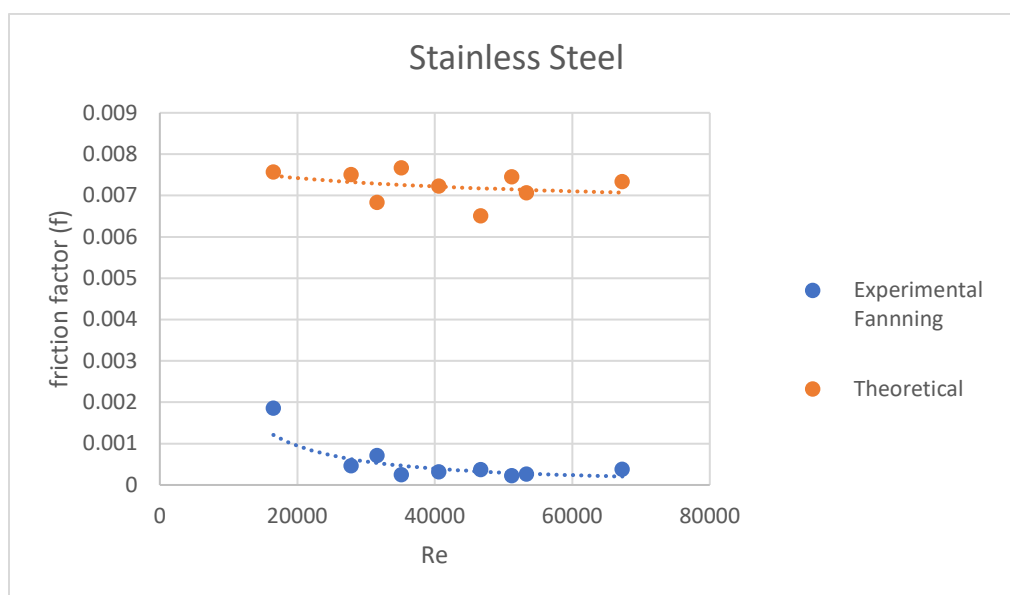


Figure 7: Reynolds number vs Fanning friction factor for stainless steel pipes comparing theoretical values obtained from the Colebrook Equation to experimental values

Table 14: Head loss for plastic pipes

PLASTIC PIPE				
Inner Diameter (ft)	Re	Head Loss Experimental (ft)	Head Loss Theoretical (ft)	% Error
0.041083333	35139.2211	0.013427419	0.12075778	88.8807004
0.051833333	27851.50483	0.00453629	0.039885334	88.6266707
0.087416667	16514.42898	0.001270161	0.00332008	61.74305542
0.041083333	67281.77671	0.041370968	0.382804907	89.19267569
0.051833333	53327.8391	0.011612903	0.126000818	90.78347001
0.087416667	31620.51088	0.001633065	0.010402385	84.30105734
0.087416667	46714.65649	0.002358871	0.020760491	88.63769199
0.041083333	51223.19787	0.026310484	0.23565057	88.83090548
0.051833333	40599.73722	0.00671371	0.077642918	91.35309454

Table 15: Head loss for stainless steel pipes

Stainless Steel Pipe				
Inner Diameter (ft)	Re	Head Loss Experimental (ft)	Head Loss Theoretical (ft)	% Error
0.041083333	35139.2211	0.005262097	0.163559217	96.78275742
0.051833333	27851.50483	0.003084677	0.050112698	93.84451939
0.087416667	16514.42898	0.000907258	0.003702828	75.49823699
0.041083333	67281.77671	0.029395161	0.573758966	94.8767404
0.051833333	53327.8391	0.006350806	0.172942442	96.32779185
0.087416667	31620.51088	0.001270161	0.0122512	89.63235171
0.087416667	46714.65649	0.001451613	0.025482634	94.30352092
0.041083333	51223.19787	0.01016129	0.337903567	96.99284313
0.051833333	40599.73722	0.00453629	0.102498842	95.5743009

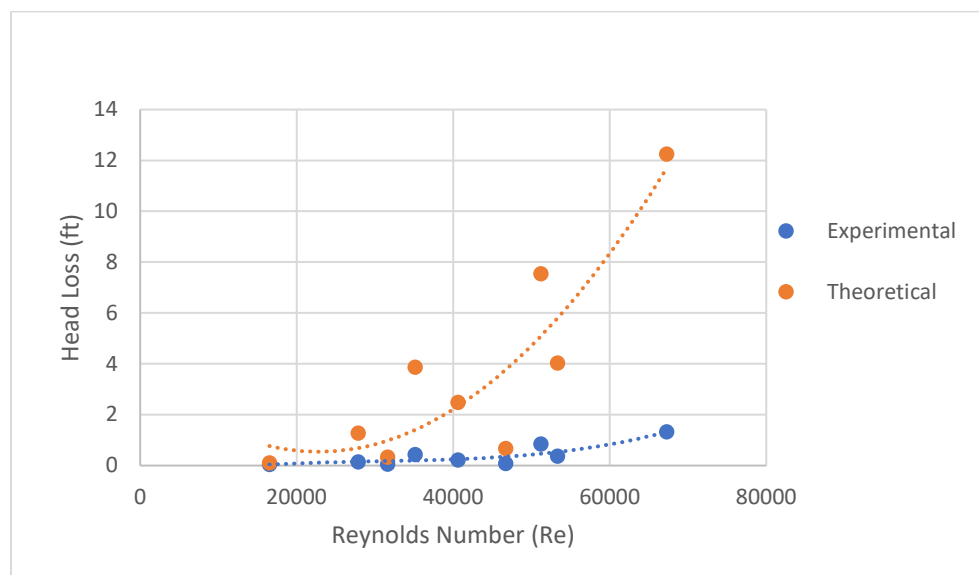


Figure 8: Plastic pipes head loss vs Reynolds number using theoretical and experimental values for comparison.

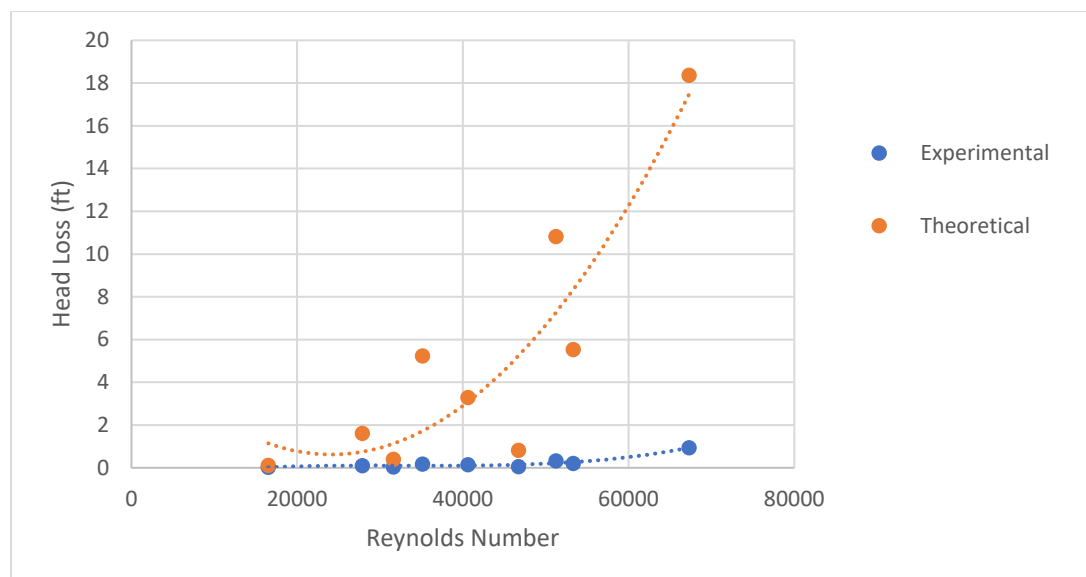


Figure 9: Stainless steel pipes head loss vs Reynolds number comparing theoretical and experimental values

Table 16: Calculation for the skin-friction loss in the straight tube

Q (SCFM)	COPPER PIPE			
	D (ft)	f Exp (Fanning)	$4*f*L/D$	V (ft/s)
10	0.0335	0.000162475	0.0776001	5.6325765
	0.0555	0.000841734	0.2426621	2.0521578
	0.085416667	0.00165185	0.3094196	0.8663861
20	0.0335	0.000602051	0.2875467	10.784808
	0.0555	0.000542681	0.1564487	3.929308
	0.085416667	0.000720908	0.1350384	1.658887
30	0.085416667	0.000536741	0.1005407	2.4507616
	0.0335	0.000209185	0.0999092	8.2107279
	0.0555	0.0006662	0.1920578	2.9914745

Appendix II: Error Analysis

There are multiple areas where errors can occur doing this experiment. First, there was a leak in the piping system which may alter the flow rate. In fact, it was noted that the rotameter tended to decrease the volumetric flow rate over time. For example, if the rotameter was set at 10 SCFM, after running the pump for a few minutes and rechecking the rotameter, the rotameter would read slightly below 10 SCFM. Air in the pipes could also cause an error in pressure drop readings. Human error can account for this by not allowing the water to flow in the pipes long enough to remove the air. The thermocouple and differential pressure transmitter accuracy at pressure

drops of less than 1 psi is not as accurate as for higher pressure drops. This may account for some variations in friction factors.

The large errors in the experimental friction factor vs the theoretical friction factors are interesting. The difference between the two values is consistently between 0.004 and 0.005. The many variations of the Colebrook equation may be part of this, as the most appropriate version of Colebrook may not have been used.

The faulty apparatus caused errors that are visible in the calculations of the pipe fitting head loss. For the head loss vs flow rate of the straight pipes, the head loss increases with the increase of the flow rate. Head loss is larger for the pipe fittings due to change of the conduits of the fluid, which is also resulting in the above experimental calculations. However, higher errors are evident in the total head loss of the system vs the flow rate because the head loss of the flow rate of 20 SCFM is higher than the head loss of the 10 SCFM and 30 SCFM: 2.15 ft, 0.77 ft, 1.07 ft, respectively.

Appendix III: Sample Calculations

Calibration of Rotameter:

Finding experimental flow rate:

$$\text{Rotameter set at 30 SCFM: } \frac{0.855 \text{ liters}}{0.036 \text{ min}} = 23.75 \text{ LPM}$$

Equation from plotting SCFM from the rotameter display to measured flow rates in LPM:

$$y = 0.7704x + 0.731$$

Calculating what 15 SCFM on rotameter is in calibrated LPM:

$$y = 0.7704 \frac{\text{LPM}}{\text{SCFM}} (15 \text{ SCFM}) + 0.731 \text{ LPM}$$

$$y = 12.287 \text{ LPM}$$

Orifice Meter:

$$\text{Flow rate LPM to m}^3/\text{s: } Q = \text{LPM} * \left(\frac{0.001 \text{ m}^3}{1 \text{ L}} \right) \left(\frac{1 \text{ minute}}{60 \text{ seconds}} \right)$$

$$Q = 8.343 \text{ LPM} * \left(\frac{0.001 \text{ m}^3}{1 \text{ L}} \right) \left(\frac{1 \text{ minute}}{60 \text{ seconds}} \right) = 0.000139 \text{ m}^3/\text{s}$$

$$\text{Differential Pressure: psi to Pa} \quad \Delta P = \text{psi} * \left(\frac{6894.76 \text{ Pa}}{1 \text{ psi}} \right)$$

$$\Delta P = 0.23 \text{ psi} * \left(\frac{6894.76 \text{ Pa}}{1 \text{ psi}} \right) = 1585.8 \text{ Pa}$$

Area of the orifice: $A_o = \pi \left(\frac{D_o}{2} \right)^2$, D_o = diameter of orifice m

$$A_o = \pi \left(\frac{0.0121412}{2} \right)^2 = 0.0001158 \text{ m}^2$$

Flow Coefficient: $C_f = \frac{Q}{A_o \sqrt{\frac{2\Delta P}{\rho}}}$, ρ = density of water kg/m³

$$C_f = \frac{0.000139}{0.000158 \sqrt{\frac{2(1585.8)}{998}}} = 0.67$$

Beta: $\beta = \frac{D_o}{D_i}$, D_i = diameter of pipe m

$$\beta = \frac{0.0121412}{0.026035} = 0.466$$

Reynolds Number: $Re = \frac{\rho D_o v}{\mu}$, v = velocity m/s, μ = viscosity of water Pa*s [4]

Mass Flowrate: $Q_{mass} = \text{flowrate} * \rho = \text{kg/s}$

$$Q_{mass} = 0.000139 * 998 = 0.1388 \text{ kg/s}$$

Straight Pipe Friction Head Loss and Friction Factors

**** 1/2 inch Copper pipe at 10 SCFM used for example calculations

Cross Sectional Area:

$$A = \pi r^2$$

r = radius of inner pipe diameter

$$A = \pi * \left(\frac{0.0335 \text{ ft}}{2} \right)^2 = 0.000881413 \text{ ft}^2$$

Velocity of water in pipe:

$$V = \frac{Q}{A}$$

Where Q = volumetric flow rate

$$\frac{0.004965 \frac{ft^3}{s}}{0.000881413 ft^2} = 5.6326 \frac{ft}{s}$$

Reynolds Number:

$$\frac{\rho V D}{\mu}$$

ρ = density of water

V = velocity

D = inner diameter of pipe

μ = viscosity of water

$$Re = \frac{62 \frac{lb}{ft^3} * 5.6326 \frac{ft}{s} * 0.0335 ft}{.000271475 \frac{lb}{ft-s}} = 43093.62$$

Fanning Friction Factor:

$$\text{Fanning } f = \frac{\Delta P D}{2 L \rho V^2}$$

$$\text{Darcy } f = \frac{\Delta P D 2}{L \rho V^2}$$

ΔP = pressure drop in pipe

L = length of pipe

ρ = density of water

V = flow velocity

$$\text{Fanning } f = \frac{76.32 \frac{lb}{ft^2} * 0.0355 ft}{2 * 4 ft * 62 \frac{lb}{ft^3} * (5.6326 \frac{ft}{s})^2} = 0.000162475$$

$$\text{Darcy } f = \frac{2 * 76.32 \frac{lb}{ft^2} * 0.0355 ft}{4 ft * 62 \frac{lb}{ft^3} * (5.6326 \frac{ft}{s})^2} = 0.000689$$

$$\frac{0.000689}{0.000162475} = 4$$

Colebrook Equation [3] to find Theoretical Fanning Friction Factor:

$$\frac{1}{\sqrt{f}} + 4 \log \left[\frac{k_s}{3.7D} + \frac{1.256}{Re \sqrt{f}} \right] = 0 \quad Re > 4000$$

f = fanning friction factor

k_s = relative roughness of pipe

D = inner diameter of pipe

Re = Reynolds number

The equation is solved using Excel Solver. An initial guess is given for f and Excel using an iterative method to solve for f so that the equation equals 0.

Initial guess of $f = 0.001$

$$\frac{1}{\sqrt{f}} + 4 \log \left[\frac{.000005 \text{ ft}}{3.7 * 0.0355 \text{ ft}} + \frac{1.256}{43093.62 \sqrt{f}} \right] = 0$$

After Excel executed What-if Analysis

$$f = 0.0055241$$

Error Analysis:

$$\frac{\text{Theoretical} - \text{Experimental}}{\text{Theoretical}} * 100 = \% \text{ Error}$$

$$\frac{.0055241 - 0.000162475}{.0055241} * 100 = 97\%$$

Head Loss:

$$h_L = f \frac{LV^2}{D^2g} \quad [3]$$

f = Fanning friction Factor

L = length of pipe

D = inner diameter of pipe

V = flow velocity

g = acceleration due to gravity

$$0.000162475 * \frac{4 \text{ ft} * \left(5.6326 \frac{\text{ft}}{\text{s}} \right)^2}{0.0335 \text{ ft} * 2 * 32.2 \frac{\text{ft}}{\text{s}^2}} = 0.009557 \text{ ft}$$

Pipe Fitting Head Loss

V- velocity, ft/s

K – Loss coefficient

Q – Flow rate, ft³/s

f – friction factor

$$V = \frac{Q}{A} = \frac{0.0049646 \frac{ft}{s}}{0.085433^2 * \frac{3.14}{4}} = 0.8660549 \frac{ft}{s}$$

$$K = \frac{\Delta P}{\rho} * \frac{2g}{V^2} = \frac{8.64 \frac{lb}{ft^2} * 2 * 32.174}{62.4 \frac{lb}{ft * s} * 0.8660} = 11.88$$

$$h_L = \left(\frac{4fL}{D} * V_{spipe}^2 + K_{elbow1} * V_{e1}^2 + K_{elbow2} * V_{e2}^2 + K_{exp} * V_{expans}^2 + K_{cont} * V_{cont}^2 \right) * \frac{1}{2g}$$

$$h_L = \left(\left(4 * 0.00165185 * \frac{4}{0.08542} \right) * 0.8664 + 11.88 * 0.866^2 + 13.86 * 0.866^2 + 69.29 * 0.866^2 + 360.560 * 0.157^2 + 2343.89 * 0.157^2 \right) * \frac{1}{(2 * 32.174)} = 2.150 ft$$

Appendix IV: Individual Team Contributions

NAME: Kellen Krupa

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Present on both days
Pre-Lab Editing		
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)	1	
Project Safety Evaluation (prelab)		
Final Lab Editing	1	Putting sections together and reviewing edit
Executive Summary (final lab)	1	Reading report and summarizing
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	6	Orifice section
Discussion (final lab)	2	Orifice section
Conclusion (final lab)		
References (final and/or prelab)	0.5	Typed citations and sources used
Data Tabulation/Graphs (final)	4	Orifice section
Error Analysis (final lab)		
Sample Calculations (final Lab)	1.5	Orifice section
Power Point Presentation	1	
Total	24	

NAME: Cheryl Slykas

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Present both lab periods
Pre-Lab Editing	1	Put together parts of prelab
Experimental Objective (Prelab)	0.5	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	

Procedure (prelab)	0.5	
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	16	Rotameter Calibration Straight Pipe Friction factors and head loss
Discussion (final lab)	1	
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	32	

NAME: Adisa Hadzic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	2	
Experimental Objective (Prelab)		
Apparatus (prelab)	2	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	10	
Discussion (final lab)	3	
Conclusion (final lab)	2	

References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)	2	
Power Point Presentation	3	
Total	31	

NAME: Marium Murtaza

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Lab operation and Pre lab
Pre-Lab Editing	2	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	3	
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)	5	Intro part of final lab
Theory (Final Lab)	5	Theory part of final lab
Results (final lab)		
Discussion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation		
Total	22	